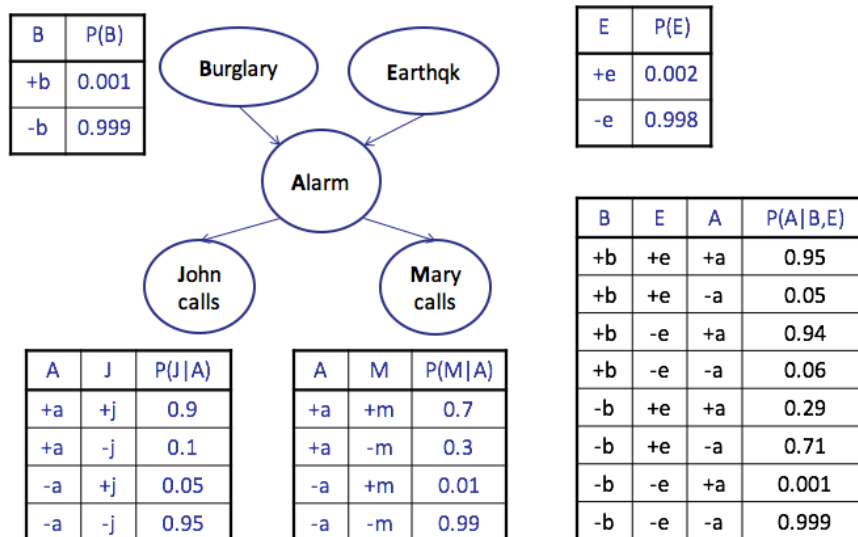


Class Test - X
CSE 317
Time: 20 minutes

Q.1. An AI-physician was asked what the odds of breast cancer would be in a woman who was initially thought to have a 2% risk of cancer but who ended up with a positive mammogram result. A mammogram accurately classifies about 70% of cancerous tumors and 80% of benign tumors. If the AI-physician used the Bayes rule to answer the queries, what would be the answer? (10)

Q.2. You are given the following Bayes Net.



Now, answer the following: (3+3+4)

- What are the independent assumptions of this Bayes Net?
- Compute $P(+b, +e, +a, +j, -m)$
- Compute $P(+j, +m)$